



COLLEGE OF ENGINEERING, PUNE

(An Autonomous Institute of Government of Maharashtra.)

END Semester Examination

(Microprocessors)

Course: S.Y. B.Tech

Branch: Computer Science and Engineering

Academic Year:

Max.Marks:60

Duration: 3 Hours

Date:

Instructions:

Student MIS No.

6 1 2 2 0 3 0 1 2

1. Figures to the right indicate the full marks.
2. Mobile phones and programmable calculators are strictly prohibited.
3. Writing anything on question paper is not allowed.
4. Exchange/Sharing of stationery, calculator etc. not allowed.
5. Write your MIS Number on Question Paper

			Marks	CO	PO
Q 1	a	Design a minimum mode CPU module around 8086 to show address bus, data bus, separate control signals for memory and I/O, power on reset and clock	4	1,2	3
	b	Interface 128 Kbyte ROM and 128 Kbyte RAM to 8086 with starting address of 00000H and 80000H respectively. Draw the memory map.	4	1	1
	c	What is string addressing mode? List all instructions used for handling string. Write program to transfer the byte string from location 20000H to 40000H onwards, number count is FFFFH.	4	2	2
Q 2	a	Explain I/O mapped I/O mode. Give examples of instructions used for accessing I/O with possible addressing modes. Explain the bus cycle execution of any one of the instructions.	4	3	1
	b	Interface 8255 to 8086 at an address F1H, give handshake signal names in the diagram for mode 1 input for port A and mode 1 output for port B. Conceptually correct diagram will earn full marks.	4	5	3
	c	Write program to demonstrate Mode 1 input and output for the design in Q.2.a above in an interrupt driven data transfer scheme. Assume use of NMI as interrupt of CPU.	4	3	4
Q 3	a	Give sequence of steps on how 8086 branches out to Interrupt Service Subroutine, if the type number of the interrupt is known. Which instruction is used to end ISS? Give sequence of steps on how it is executed	4	4	2
	b	Interface master 8259 to 8086 with base address of F0H, IR2 of the master are connected with slave 8259 with base address of F1H. Design for buffered cascaded mode.	5	5	3
	c	For the design in question 3.A, Initialise 8259 for base type number for Master as 40H and for slave it is 80H. with IR1 of	5	3	4

		master has vector location address 8000:6000H. IR3 of slave has vector location address 8000:7000H. Write Interrupt Service Subroutine for Master 8259's IR1 level, to decrement contents of memory pointed by BP by one. Write Interrupt Service Subroutine for IR3 of slave to increment contents of memory location pointed by BP by one. Ensure normal fully nested mode.			
Q4	a	Interface scanned key matrix and two seven segment displays in decoded mode for 8279 at port address of C1H. Assume digit and segment selection lines of seven segment displays are properly buffered. Use NMI interrupt of CPU for an Interrupt driven mode for 8279.	4	5	3
	b	Write program to initialize 8279 interface design of Q4.a above and read the contents of FIFO RAM to store it from memory locations pointed by BX register onwards, in an interrupt driven mode as designed.	4	3	3
	c	What is DMA? Explain the function of HOLD and HLDA signals of 8086	2	5	1
Q5	a	Explain following signals and prefixes for coprocessor configuration in maximum mode of 8086, RQ/GT0, RQ/GT1 QS0, QS1 TEST BUSY ESC LOCK WAIT	6	6	4
	b	Explain the execution details and number of bus cycles required AX = 123BH, CS=3000H, IP=0000H, BP=2000H, SP=2000 SS=1000H CS:IP OPCODE MNEMONICS 3000:0001H 2E 89 5E FD CS: MOV [BP-03H], AX	2	3	1
	c	Draw architecture of 8086. List addressing modes of 8086 with one example each	4	4	1

8259 Data:

ICW1: A0=0

X	X	X	1	LTIM	ADI	SNGL	IC4
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ICW2: A0=1

T7	T6	T5	T4	T3	X	X	X
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ICW3 (MASTER): A0=1

S7	S6	S5	S4	S3	S2	S1	S0
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ICW3 (SLAVE): A0=1

0	0	0	0	0	ID2	ID1	ID0
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ICW4: (A0=1)

0	0	0	SFNM	BUF	M/S	AEOI	μPM
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OCW1: (A0=1)

M7	M6	M5	M4	M3	M2	M1	M0
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OCW2: (A0=0)

R	SL	EOI	0	0	L2	L1	L0
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R	SL	EOI	
0	0	1	NON SPECIFIC EOI
0	1	1	SPECIFIC EOI
1	0	1	ROTATE ON NON SPECIFIC EOI
1	0	0	ROTATE IN AUTOMATIC EOI (SET)
0	0	0	ROTATE IN AUTOMATIC EOI (CLEAR)
1	1	1	ROTATE ON SPECIFIC EOI
1	1	0	SET PRIORITY

8279 Data:

KEYBOARD/DISPLY MODE SET

0	0	0	D	D	K	K	K
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DD	
00	8 CHARACTER DISPLAY LEFT ENTRY
01	16 CGARECTER DISPLAY LEFT ENTRY
10	8 CHARACTER DISPLAY RIGHT ENTRY
11	16 CHARACTER DISPLAY RIGHT ENTRY

KKK	
000	ENCODED SCAN KEYMATRIX 2 KEY LOCKOUT
001	DECODED SCAN KEYMATRIX 2 KEY LOCKOUT
010	ENCODED SCAN KEYMATRIX N KEY ROLLOVER
011	DECODED SCAN KEYMATRIX N KEY ROLLOVER
100	ENCODED SCAN SENSOR MATRIX
101	DECODED SCAN SENSOR MATRIX
110	STROBED INPUT ENCODED DISPLAY SCAN
111	STROBED INPUT DECODED DISPLAY SCAN

Program Clock:

0	0	1	P	P	P	P
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Read FIFO/Sensor RAM:

0	0	0	A	I	X	A	A	A
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FIFO Status Word:

DU	S/E	O	U	F	N	N	N
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